

A FIELD GUIDE TO THE BUYING BRAIN



The Neuro-Web Design Playbook

How the Brain Buys

*Thirty evidence-graded laws of attention, cognitive load, persuasion,
pricing, trust, memory, color and conversion — with the myths
stripped out and the mechanisms drawn.*

BUILT FOR YUSUF GADELRAH · JULY 2026 · 8 RESEARCH
DOMAINS · 24 FIGURES · 30 LAWS

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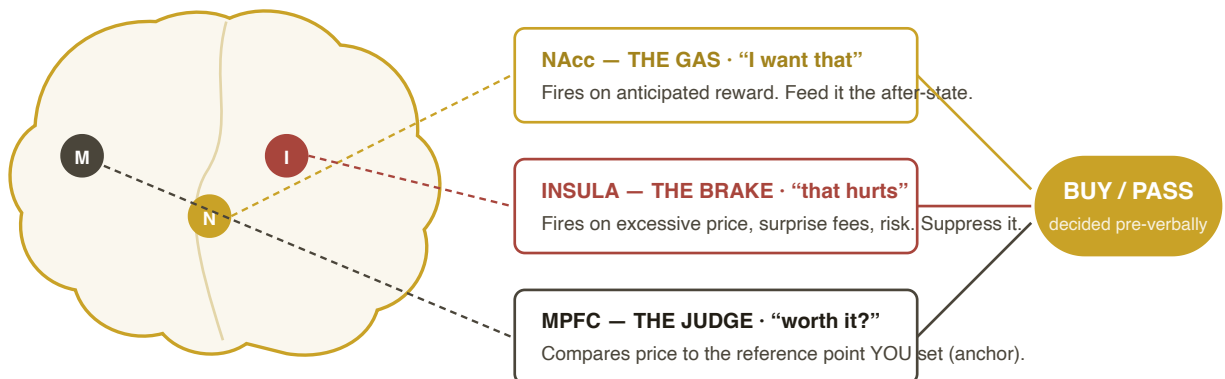
Every principle here is graded against primary sources. Where the popular version of a claim failed replication — the jam study, the 7 ± 2 menu rule, red buttons, the goldfish attention span, oxytocin storytelling — it is flagged in a myth table, not repeated. Build on the laws; quote the flags.

The Buying Circuit

One model explains almost everything in this book: buying is a race between a gas pedal, a brake, and a judge.

In 2007, Knutson, Rick, Wimmer, Prelec and Loewenstein put people in an fMRI scanner, showed them a product, then its price, and watched them decide (*Neuron*, n=26). Three signals predicted the purchase **seconds before the choice was made** — and better than what people said they wanted:

- **Nucleus accumbens (NAcc)** fired when the product was desirable — the **gas pedal**. It responds to *anticipated* reward, not the reward itself.
- **Anterior insula** fired when the price felt excessive — the **brake**. The same region processes pain and disgust: hence "the pain of paying."
- **Medial prefrontal cortex (MPFC)** integrated the two into a net "worth it?" signal — the **judge**. It tracks price *relative to a reference point*, which is why anchoring works and why discounts activate it.



Knutson, Rick, Wimmer, Prelec & Loewenstein — "Neural Predictors of Purchases," *Neuron* 2007 (n=26). Each signal predicted purchase beyond

Figure 0 — The buying circuit. Every tactic in this book either presses the gas (anticipated reward), eases the brake (pain, risk, surprise cost), or briefs the judge (reference points). There is no "buy button" — only this net-value race, run before words.

How to use the model. When auditing any page, ask three questions in order: *What does the gas pedal see first?* (the outcome, not the feature list) · *Where does the brake spike?* (surprise fees, risky forms, unclear totals, walls of choice) · *What reference point does the judge get?* (an

anchor you chose, or one the visitor invents). The eight chapters that follow are the attention system that feeds this circuit, the working-memory bottleneck it runs through, and the memory system that decides whether the visitor ever comes back.

△ Honesty note: insula activation is not literal pain, and reverse inference has limits. The neuro layer explains *why* the behavioral evidence looks the way it does; the field experiments and A/B data are what justify shipping.

Attention & Salience — Winning the Glance

Users don't read pages; they forage them. Your job is to control where the eye lands in the first three seconds — because salience is zero-sum.

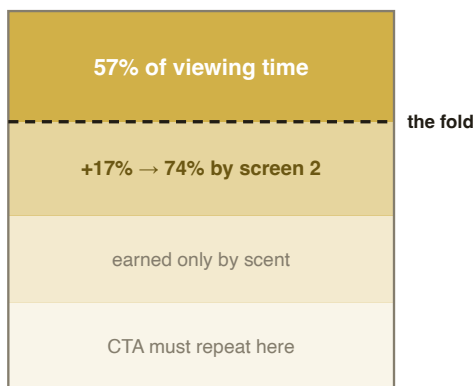
1.1 The fold is a gradient, not a wall

MECHANISM — Information-foraging theory (Pirolli & Card, Xerox PARC): scrolling is a *cost* users only pay when the visible band promises payoff ("information scent"). Attention decays smoothly down the page — it doesn't stop at the fold, it discounts below it.

DO — Value proposition + one CTA inside the first screenful; let the next section peek 40–80 px above the fold (no "false floor" — never end the hero on a clean full-viewport edge); repeat the *same* CTA at every decision point down the page. Don't cram everything above the fold — that kills the scent gradient and raises complexity (Ch. 2).

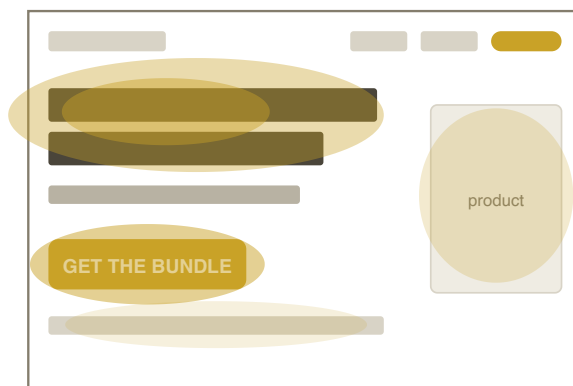
EVIDENCE — NN/g 2018 eyetracking, 130,000+ fixations: **57% of viewing time above the fold, 74% within the first two screenfuls**; the 100 px just above the fold get **102% more views** than the 100 px below it. Google viewability: 56.1% of display impressions are never seen at all.

A · THE ATTENTION GRADIENT



+102% views just above vs. just below the fold (NN/g)

B · WHERE THE HEAT MUST LAND



Pass condition: top-3 hottest regions = headline, primary CTA, product/pro

Fail: a stock photo, cookie banner or logo wins the heat.

Figure 1 — The fold gradient and the target heat signature. Attention is front-loaded (57% above the fold) but continues wherever scent is maintained. Pre-test the hero with a 5-second test or saliency model: if the predicted heat lands anywhere but headline / CTA / proof, redesign before shipping.

1.2 They scan — the F-pattern is a symptom of your failure, not a template

MECHANISM — Only $\sim 2^\circ$ of foveal vision is sharp; reading is a saccade/fixation loop steered by cues. With no cues (headings, bolding, contrast), the eye defaults to the cheapest sweep: two horizontal passes + a left-edge scan — the F. NN/g is explicit: the F appears **when formatting gives the eye nothing to aim at**. Motivated readers with good cues show the **layer-cake** pattern instead — hopping heading to heading.

DO — A descriptive subhead every 2–3 paragraphs; front-load the first two words of every heading, link and bullet; bold the key term only; left-align body text; cut words ruthlessly (each 100 extra words buys only ~ 4.4 s of attention).

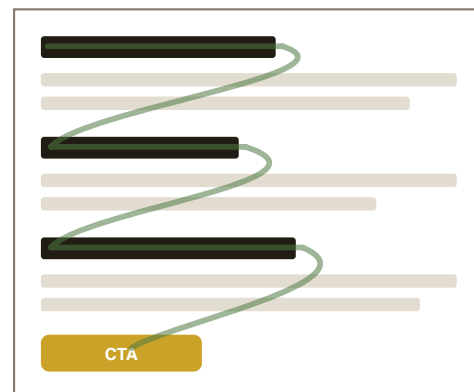
EVIDENCE — Nielsen 2006 (232 users), refined 2017; Weinreich et al. 2008 (45,237 page views): users read **at most 28% of the words, more likely $\sim 20\%$** .

✗ NO CUES → F-PATTERN (failure)



Everything right of the F stem is never seen.

✓ SUBHEADS → LAYER-CAKE (engineered)



Front-loaded subheads carry the scan straight to the CTA.

Figure 2 — F-pattern vs. layer-cake. The F is what eyes do on a wall of undifferentiated text; the layer-cake is what they do when subheads exist every 2–3 paragraphs. You choose which one your page gets. (Note: the "Z-pattern" beloved of design blogs appears in no eyetracking taxonomy — it's newspaper lore, usable as a composition heuristic for sparse heroes, never citable as science.)

1.3 One winner per screen — salience is computed winner-take-all

MECHANISM — Bottom-up attention runs on a saliency map (Koch & Ullman 1985; Itti, Koch & Niebur 1998): feature-contrast maps for luminance, color and orientation sum into one map, a **winner-take-all** network picks the single peak, then inhibition-of-return moves on. Treisman's feature-integration: a one-feature difference pops out in parallel; competing differences force slow serial search. Von Restorff (1933): the isolate is what's remembered — and **two isolates are zero isolates**.

DO — Reserve one accent hue for primary actions and use it *nowhere else*; one gold button per screenful, everything else ghost weight; whitespace is literally center-surround contrast — give

the CTA room; minimum 3:1 non-text contrast; on dedicated landing pages strip global nav (competing exits have salience too). Run the squint test: blur the page — if the CTA isn't the first thing you see, it's just colorful, not salient.

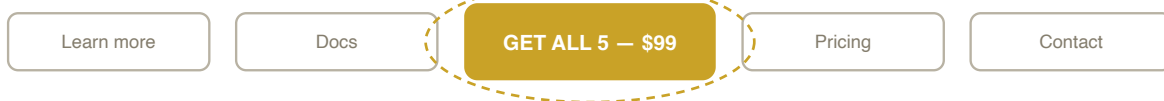
EVIDENCE — Deep saliency models (MIT300 benchmark) predict human fixations from feature contrast alone. The famous HubSpot "red beat green +21%" test was an **isolation** win — red against a green-dominant page — not a red win.

✗ FIVE LOUD ELEMENTS — NO PEAK, SERIAL SEARCH



Saliency map output: five equal peaks → attention serializes → decision cost → exit.

✓ ONE RESERVED ACCENT — INSTANT WINNER



Winner-take-all resolves in one glance. The Von Restorff isolate is also what gets REMEMBERED after the tab closes.

Figure 3 — Salience is zero-sum. The brain's winner-take-all network can crown exactly one peak per glance. Every additional loud element is a tax on the one that pays your rent.

1.4 Don't look like an ad — and spend motion like money

MECHANISM — Banner blindness is *learned suppression*: years of exposure built a schema for "ad" (top strip, right rail, IAB sizes, animation, stock face + bold offer) that users top-down inhibit — NN/g documents "hot potato" scanning where eyes actively bounce away. The schema misfires on *your own content* when it shares those features. Motion: abrupt onsets capture attention near-obligatorily (Yantis & Jonides) — but only when **rare** (Folk & Remington 2015). A looping animation burns its capture in cycle one, then habituates into background noise while degrading reading of adjacent text and tripping the ad schema.

DO — Conversion content in the main column, never in a right rail or a strip pinned to the top; never size promos at IAB dimensions (300×250, 728×90); style offers as native content. Motion: one entry animation (200–300 ms) on the one element you want noticed, no loops, **zero auto-rotating hero carousels**, honor prefers–reduced–motion. If a face appears on the page, use **averted gaze aimed at the headline or CTA** — mutual gaze pools fixations on the face and steals them from your offer (Sajjacholapunt & Ball 2014: averted gaze ≈ 5× product dwell and ~3× brand recognition vs. no face).

EVIDENCE — Benway & Lane 1998: users missed banner-styled links *containing the answer they were told to find*. NN/g: a right rail took 25% of content area and got **0.8% of fixations**. Notre

Dame carousel data: ~1% of visitors clicked any slide, **84% of those clicks on slide 1** — slides 2–5 effectively don't exist. Adobe client: removing the slider = **+23% sales**.

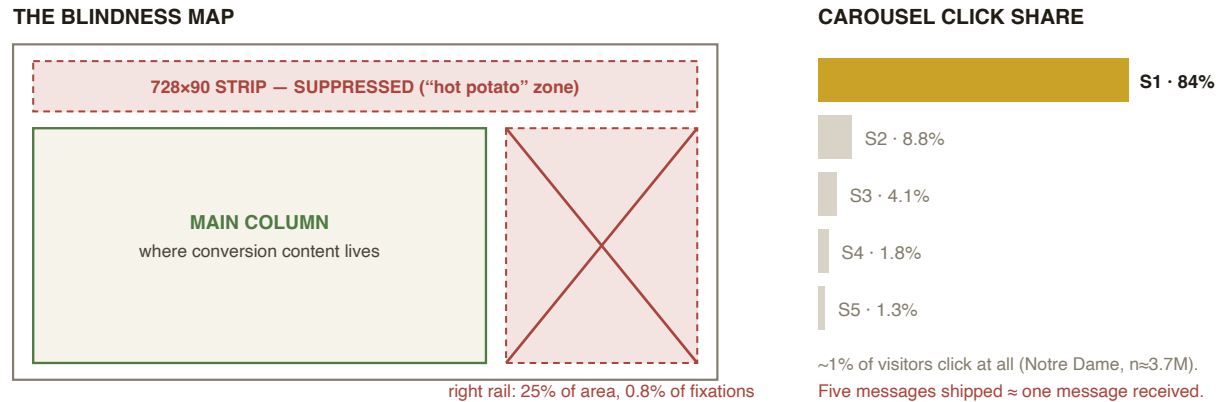


Figure 4 — Banner blindness and the carousel tax. Anything shaped, placed, or animated like an ad inherits the ad's invisibility. A rotating hero is the worst of all worlds: it looks like an ad, moves like an ad, and gives slides 2–5 a rounding-error audience.

ATTENTION MYTH	REALITY
"Design the page as an F / Z"	✗ The F is a diagnosis of missing cues; the Z has no eyetracking basis. Engineer the layer-cake.
"Users don't scroll"	✗ 57% above fold / 74% in two screens — a gradient, not a wall. Long pages convert when scent holds.
"Red (or green, or orange) buttons convert best"	✗ Only local contrast + isolation generalize. Hue claims are untransferable.
"8-second attention span, less than a goldfish"	✗ Fabricated statistic with no primary source. Never cite it.
"AI heatmaps are 93–96% accurate"	✗ Vendor marketing. Saliency models are free-viewing triage tools, not evidence.

SHIP LIST — ATTENTION

- Value prop + one CTA in viewport one; next section peeks over the fold; same CTA repeated down-page.
- Subhead every 2–3 paragraphs, first two words load-bearing; bold key terms only.
- One reserved accent color; one gold button per screen; squint-test the hero.
- Nothing ad-shaped, ad-placed, or looping; faces gaze at the CTA; prefers–reduced–motion honored.

Cognitive Load — The Four-Chunk Bottleneck

Working memory holds ~4 chunks (Cowan 2001, not Miller's 7). Every ounce of effort spent decoding your page is stolen from the decision to buy.

Sweller's Cognitive Load Theory splits mental effort into **intrinsic** load (the decision itself), **extraneous** load (imposed by bad design), and **germane** load (useful processing). You can't reduce the first; your entire job is deleting the second. The visitor's aesthetic verdict on your complexity is formed in 17–50 ms — before a single word is read (Lindgaard 2006; Tuch/Google 2012: **low visual complexity + high prototypicality** wins).

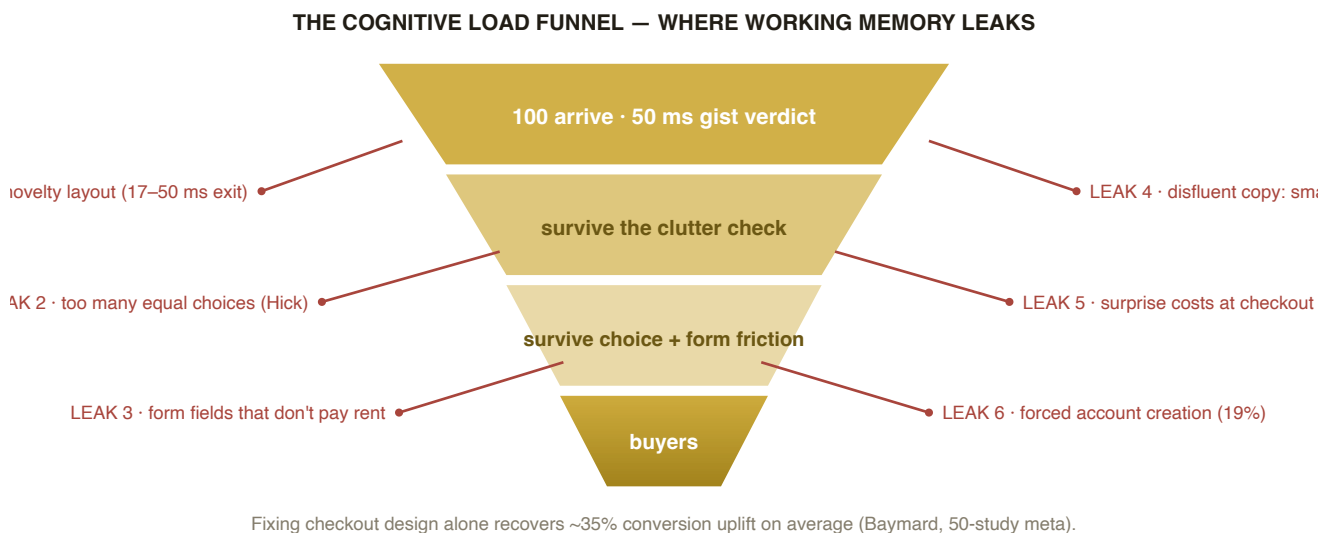


Figure 5 — The cognitive load funnel. Visitors don't leave because they decided against you — they leave because deciding got expensive. Each leak is extraneous load with a named fix in this chapter.

2.1 Hick's Law — fewer, clearer options at every decision point

MECHANISM — Decision time grows logarithmically with the number of comparable options: $RT = a + b \cdot \log_2(N+1)$ (Hick 1952; Hyman 1953). Boundary condition: it applies to *unfamiliar*, *comparable* choices — a user hunting a known target scans, so it never justifies hiding known links.

DO — One primary CTA + at most one ghost secondary; 3 pricing tiers with one preselected "Most Popular" (a default collapses N toward 1); never offer two near-identical actions ("Sign up" vs "Get started"); trim nav to what maps to visitor goals — 4 links beats 7 on a selling page.

EVIDENCE — Among the most replicated laws in psychophysics; applied: an 8-question signup cut to 3 reduced completion time $\sim 1/3$ and raised signups. Unbounce (41k pages): 1-CTA pages convert 13.5% vs 10.5% for 3+ CTAs.

2.2 Fitts's Law — make the money button big, close, and thumb-reachable

MECHANISM — Time to hit a target: $T = a + b \cdot \log_2(2D/W)$ — rises with distance, falls with size (Fitts 1954). Motor difficulty is felt as friction even when users can't name it. Screen-edge "infinite targets" are a cursor phenomenon only — the advantage does **not** exist on touch.

DO — Touch targets $\geq 44 \times 44$ pt (Apple) / 48 dp (Material); submit button directly after the last field, never far away; primary mobile CTA in the bottom thumb zone (or a sticky bottom bar); icon+label = bigger effective target; space targets so mis-taps don't punish.

EVIDENCE — NN/g Fitts applications; SaaS case: moving Save adjacent to the form + enlarging to 56 px roughly halved completion time. Thumb-zone tap accuracy $\sim 96\%$ vs $\sim 61\%$ in the stretch zone.

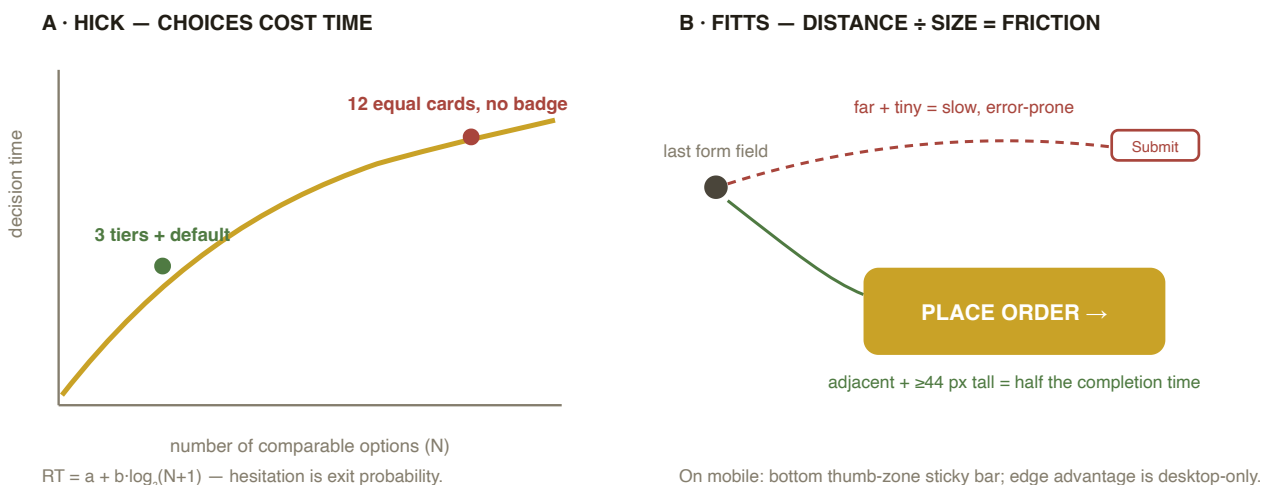


Figure 6 — The two friction laws. Hick taxes the choosing; Fitts taxes the doing. Both are paid in milliseconds that feel like doubt.

2.3 Every form field must pay rent

MECHANISM — Each field adds interaction cost, error probability, a privacy micro-decision, and a risk cue. Ambiguous fields cause silent task failure — users stop without telling you why.

DO — Single "Full Name" field; hide "Address Line 2" and coupon codes behind links (a visible coupon field sends buyers off-site hunting for codes); billing = shipping by default; browser autofill attributes; inline validation; guest checkout default with account creation offered *after* the order.

EVIDENCE — Baymard: average checkout shows 11.3 fields, ~8 needed; 30% of test users came to a full stop at "Address Line 2." Expedia deleted one optional "Company" field → +\$12M/year. Venture Harbour: one long form → multi-step: 0.96% → 8.1% conversion.

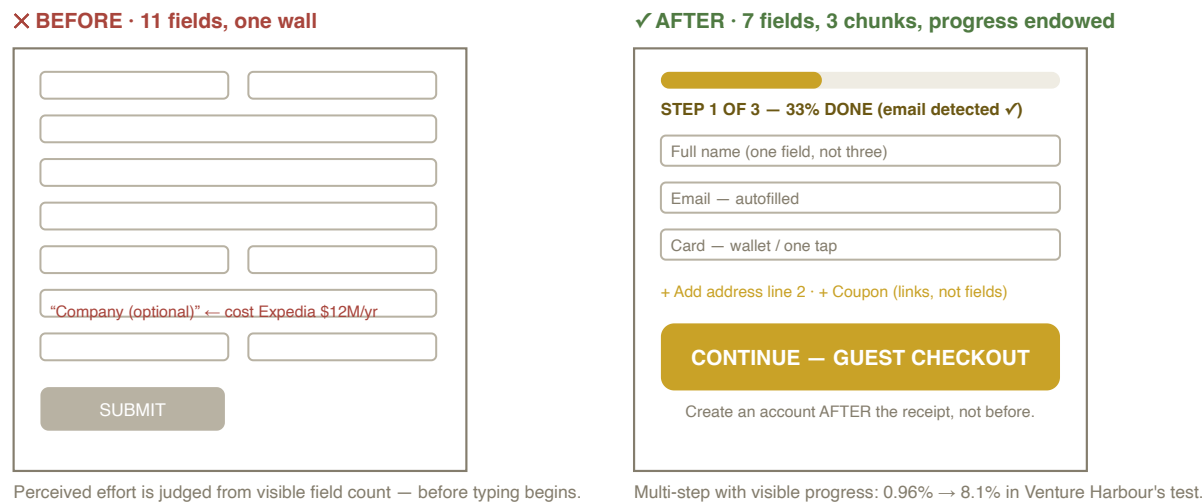


Figure 7 — The form audit. Chunking (Cowan's real 4±1 limit), endowed progress, adjacent submit, and deleted fields — the fastest measurable wins in CRO, worth eight figures at Expedia's scale.

LOAD MYTH	REALITY
"Menus must have 7±2 items" (Miller)	✗ Recall limit, not recognition. Visible menus have no such cap — chunk under labels instead; true WM limit ≈ 4 (Cowan 2001).
"Fewer products always sell more" (jam study)	✗ Meta-analytic mean ≈ 0 (Scheibehenne 2010). Overload appears only with comparable options + uncertainty (Chernev 2015): fix with filters, defaults, "Most Popular" — not amputation.
"The 3-click rule"	✗ No correlation between clicks and success. Count thinking-per-click, not clicks.
"Edge targets are fastest" on mobile	✗ Cursor-only phenomenon. On touch, edges can be slower.

SHIP LIST — COGNITIVE LOAD

- One page, one job, one primary CTA; nav $\leq$ 4–5 goal-mapped links.
- 3 tiers + preselected default wherever a choice is offered; never twin CTAs.
- ≥ 44 px targets; submit adjacent to last field; sticky thumb-zone CTA on mobile.
- Delete every field that doesn't pay rent; guest checkout default; progress bars start part-filled.
- Pre-computed arithmetic (\$29/mo, not $\$348/\text{yr} \div 12$); 16 px+ high-contrast type everywhere money moves.

Persuasion & the Dopamine Loop — Selling the After-State

Dopamine is the chemistry of wanting, not liking. People don't buy products; they buy the anticipated version of themselves that already has the outcome.

3.1 Sell the anticipation, not the object

MECHANISM — Midbrain dopamine neurons fire a *reward prediction error* (Schultz): once a cue reliably predicts reward, the burst **migrates backwards from the reward to the cue**.

Knutson 2001: NAcc scales with *anticipated* magnitude; the outcome itself is processed elsewhere. Berridge & Robinson's dissociation: dopamine-depleted rats still *enjoy* sugar but won't work for it — dopamine drives **wanting** (incentive salience), not pleasure.

DO — Above the fold, show the **after-state**: the inbox already answered, the invoice already paid, the report already generated. Ranked by anticipatory power: interactive demo > live preview > screenshot > feature bullets. Reuse one consistent cue (same button color, same phrasing, same micro-animation) at every entry point so the anticipatory response transfers to it. Payoff first, price second — never let the CTA be the first mention of the reward.

EVIDENCE — Knutson et al. 2001, 2007; in practice, interactive product demos and free-tier "aha" flows consistently beat static feature pages because they *manufacture* the anticipatory state.

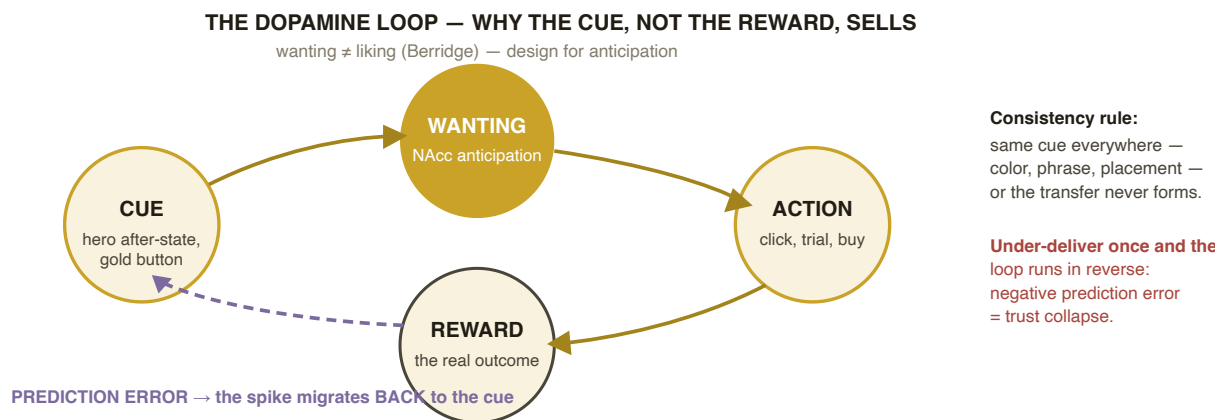


Figure 8 — The dopamine anticipation loop. After one good experience, the gold button itself fires the wanting. This is also the argument for congruence: an over-promising page trains a negative prediction error instead.

3.2 Endowed progress and the goal gradient

MECHANISM — Effort accelerates as perceived distance to a goal shrinks (Hull 1932); Nunes & Drèze 2006 showed the reference point is manipulable — grant artificial head-start progress and people behave as if genuinely closer, and finish faster (Kivetz 2006 replicated with café cards).

DO — Never show an empty progress bar. "Step 1 of 4" renders as "2 of 5 complete — we filled in your details." Pre-tick 1–2 onboarding items ("✅ Account created"). Prefill everything inferable — each prefilled field is friction removed *and* progress granted. Multi-step only if progress is visible.

EVIDENCE — Car-wash study: 8-stamp card from zero = 19% completion; 10-stamp card with 2 free stamps (identical work) = **34%** — near-doubling from pure framing.

3.3 Defaults beat persuasion

MECHANISM — Defaults stack three forces: effort cost, implied endorsement, and status-quo reference points. This is choice architecture, not rhetoric — the highest-leverage, lowest-cost intervention in the behavioral toolkit.

DO — Guest checkout as the default path; preselect the plan you want sold (with terms fully visible); default the least-commitment next step that captures intent. Legit: preselecting a visible plan. Not legit (and now enforced against): pre-checked paid add-ons and hard-to-cancel subscriptions (ROSCA/FTC).

EVIDENCE — Johnson & Goldstein 2003 (*Science*): organ-donor consent 42% opt-in vs **82% opt-out** — same people, same question, different default. Baymard: 19% of abandoners cite forced account creation.

A · ENDOWED PROGRESS (Nunes & Drèze)

"0 of 8" — cold start

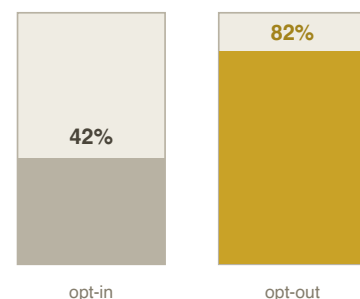


"2 of 10 already done" — same real work



Web forms: "Step 1 of 3 — 33% done", pre-ticked onboarding items, prefilled fields. The goal gradient does the pulling.

B · THE DEFAULT FLIP



Johnson & Goldstein 2003 — same question, different default. Preselect the good choice.

Figure 9 — Free conversions from framing. Two of the largest effects in this book cost zero pixels of new design: grant progress before asking for effort, and make the desired path the default path.

3.4 Loss framing, honest scarcity, and the reactance line

MECHANISM — Losses loom larger than gains (prospect theory) — but Gal & Rucker 2018 showed the universal "2×" version is dead; the asymmetry is reliable mainly for concrete, owned things at real stakes. Scarcity raises value (commodity theory) because it collapses "decide later" — but Brehm's **psychological reactance** means *detected* manipulation pushes back hard, and modern users are calibrated to fake timers.

DO — Loss-frame the **inaction**, never the purchase ("at 200 loads/month, manual AR costs you ~\$1,900/mo"), and create real endowment before the ask (trial with their data in it). Scarcity only when provable on demand: real caps ("8 pilot slots, 3 taken"), dated price changes that actually happen ("launch pricing ends Aug 15 — then it ends"). Give every deadline a reason. No fake counters, no "17 people viewing."

EVIDENCE — FTC dark-patterns report (2022) names fake timers and false scarcity as enforceable deception; Booking.com's countdown practices drew multi-million-euro fines; Amazon paid **\$2.5B** (2025) over Prime dark patterns. Honest deadline lifts are real; fake ones are now a legal liability, not a hack.

PERSUASION MYTH

REALITY

"Dopamine = pleasure chemical"

✗ Dopamine = wanting/anticipation (Berridge). Design for the craving state, not the "hit."

"95% of decisions are subconscious"

✗ No source exists; misattributed to Zaltman. Unmeasurable as stated.

"Sell to the reptilian brain"

✗ MacLean's triune-brain model is rejected by modern neuroscience. Dead framework.

"Losses hurt exactly 2× as much"

✗ Contingent, not constant. Expect single-digit lifts from framing swaps; A/B it.

"Subliminal priming sells Coke"

✗ Vicary admitted fabricating the 1957 study. Real subliminal effects are tiny and need an active goal.

SHIP LIST — PERSUASION

- Hero shows the after-state; add an interactive or live preview wherever possible.
- One consistent CTA cue (color + phrase) across every page and email.
- Progress bars pre-filled; onboarding pre-ticked; everything inferable prefilled.
- Guest checkout and the recommended tier preselected — visibly, honestly.
- Loss-frame inaction with the visitor's own numbers; scarcity only with receipts.

Neuro-Pricing — Anchors, Decoys, and the Pain of Paying

A price is never judged alone. The brain computes it against whatever reference point arrived first — so arrive first.

4.1 Value before price — control which question the judge asks

MECHANISM — Karmarkar, Shiv & Knutson 2015 (*JMR*): information *order* changes the MPFC's computation. Product-first → "Do I like it?" (desire-driven). Price-first → "Is it worth it?" (cost filtering — desire never engages). Exception: lead with the number when you're the obviously-cheap utilitarian option ("same job, 70% less").

DO — On every pricing card, DOM order = outcome headline → one-line proof → features → **then** the number. Never open a landing page with a bare price table for a desire product.

4.2 Anchor with a real, relevant, precise comparison

MECHANISM — Anchoring-and-adjustment (Tversky & Kahneman 1974) is one of the few classic effects that replicated cleanly in Many Labs. Precision sharpens it: precise anchors (\$4,998) sit on a finer mental scale, so people adjust less from them than from round ones (Janiszewski & Uy 2008). But **arbitrary** anchors are dead: the famous SSN-digit result failed a well-powered 2025 replication (~9× smaller, insignificant). Relevant comparisons move money; random numbers don't.

DO — Put the alternative's true cost adjacent to your price ("an AR clerk runs \$1,800–2,200/mo — FreightDesk is \$249"). Anchor and offer in *one visual frame* (~~\$285~~ **\$99**), never separate scrolls. Strike-throughs must be genuine prices (FTC/EU enforcement is active). For negotiated B2B, quote precise figures (\$1,847) — round numbers invite round counteroffers.

SAME PRICE, TWO BRAINS — THE REFERENCE POINT DOES THE SELLING



SERVICE ANCHOR — CHANGE THE COMPARISON CLASS



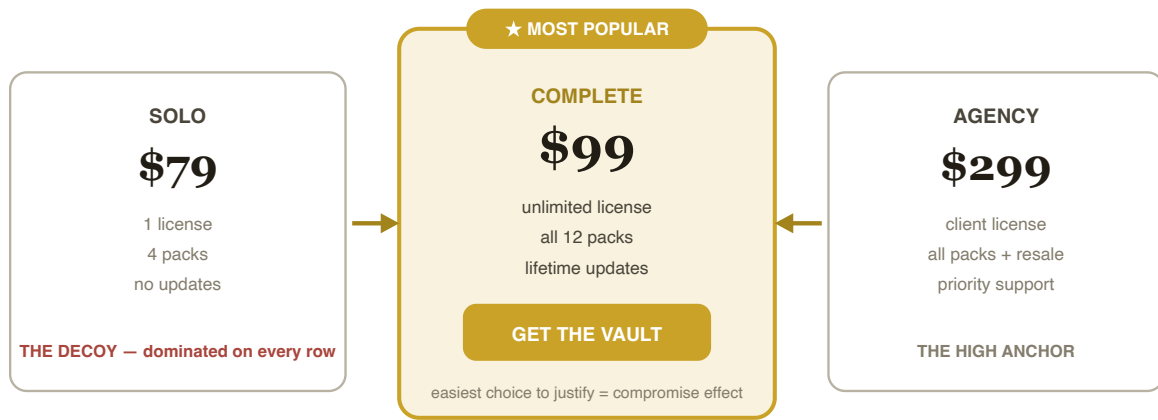
The comparison class is now labor, not software — and \$8/day retrieves "coffee," not "rent" (Gourville's pennies-a-day, 1998).

Figure 10 — Pricing anchors. The judge (MPFC) computes price – reference. Supply the reference: a genuine strike-through, an itemized value stack, the alternative's payroll cost, and a per-day reframe with the real total still dominant.

4.3 Three tiers: compromise first, decoy second — with a numeric grid

MECHANISM — Two stackable effects. **Compromise/extremeness aversion** (Simonson 1989): the middle is the easiest choice to justify. **Asymmetric dominance** (Huber, Payne & Puto 1982): an option clearly beaten by your target on every attribute lifts the target's share — *The Economist's* useless print-only tier moved bundle choice 32% → 84% (Ariely's classroom demo). Sober flag: Frederick et al. 2014 got only 11 reliable decoy effects in 91 attempts — the effect needs **comparable numeric attributes**. A pricing table is one of the few real-world contexts that qualifies.

DO — Exactly 3 tiers, target in the middle, badged "Most Popular"; render a numeric attribute grid (seats, docs, licenses, turnaround); the decoy must be dominated (same-or-higher price, strictly fewer numbers); a 4th "Contact us" enterprise tier is an anchor, not a decoy — fine, but keep the grid intact.



Both neighbors exist to make the middle inevitable. Desktop order: consider most-expensive-first so the high number is encoded first.

Figure 11 — The engineered middle. Compromise pulls to the center; the dominated decoy and the high anchor push from both sides. Requires a visible numeric grid — the decoy effect evaporates on vibes-based options.

4.4 Engineer down the pain of paying

MECHANISM — Prelec & Loewenstein's coupling hypothesis (1998): the tighter payment is coupled to consumption in time and salience, the more it hurts (the ticking taxi meter). Decoupled instruments hurt less (credit > card-on-file > wallet one-tap). Flat rates are systematically over-chosen and users are *happier* despite overpaying (Lambrecht & Skiera 2006) — metered pricing makes every use an insula event.

DO — Flat monthly/annual over metered wherever unit economics allow; one currency symbol per card, calm price typography; wallet checkout (Apple Pay / Link); decouple timing ("\$0 today — charged in 14 days"); **integrate losses, segregate gains** (Thaler): one price, many itemized value lines — never "+\$29 onboarding, +\$19 support" add-ons at the decision point.

EVIDENCE — Knutson 2007 (insula predicts non-purchase); Prelec & Simester 2001 (credit-card WTP $\approx 2\times$ cash — ceiling estimate; 2024 meta-analysis: reliable but moderate); Morwitz 1998 + StubHub field data show partitioned fees work by exploiting recall failure — which is why the FTC junk-fees rule (May 2025) and CA SB 478 now ban hiding them. Show the all-in total; partition the *value*, never the fees.

PAYMENT COUPLING — WHEN THE INSULA FIRES

✗ **METERED / DRIP** — pain at every use



✓ **FLAT / PREPAID** — pain once, far from use



Corollary: a fee revealed at checkout re-couples payment at the worst moment — #1 abandonment cause, 39–40% (Baymard), and now an FTC target.

Figure 12 — Coupling. Same total dollars, different pain schedule. Flat and prepaid pricing concentrates the insula event once; metered pricing and dripped fees fire it forever.

4.5 Justify the price; format it for the buying mode

MECHANISM — Cost transparency acts like interpersonal self-disclosure: it raises attraction and purchase (Mohan, Buell & John 2020 — +21% soup field experiment, +22% wallet sales over 92 days). Unexplained discounts trigger quality suspicion (wholesaler field experiment, n=780: a naked discount *reduced* purchase). Price format: round prices feel right for hedonic buys, precise prices read as computed for utilitarian buys (Wadhwa & Zhang 2015); charm endings only matter when the left digit drops (\$30→\$29 yes, \$3.60→\$3.59 no) and can cheapen premium positioning.

🟡 — Add a short "why it costs this" block (your math, voluntarily shown). Every discount gets a named, dated reason ("launch pricing through Aug 15," "founding-client rate — first 10 pilots"). Templates/luxury: round numbers. Ops tools/B2B ROI: precise numbers. Never spend a discount that doesn't move the left digit.

PRICING MYTH

REALITY

"Random anchors move willingness-to-pay"	✗ Failed high-powered replication (2025). Only relevant, comparable anchors work.
"The decoy effect works on anything"	✗ 11/91 attempts outside numeric grids. Works on pricing tables; not on vibes.
"\$X.99 always beats \$X.00"	✗ Modest, conditional; fires only on a left-digit change; hurts premium brands; 2026 preregistered N=729: no purchase-intent effect.
"Hide fees until checkout — it converts"	✗ Empirically true, legally radioactive (FTC junk-fees rule, CA SB 478) and trust-destroying. Total price upfront.
"Removing the \$ sign raises spend"	✗ One restaurant study (n=201, +8.15%), never web-replicated. Test, don't assert.

SHIP LIST — PRICING

- Reorder every card: outcome → proof → features → price (price last in the DOM).
- Real anchor beside every price; itemized value stack summing above the bundle price.
- 3 tiers + numeric grid + "Most Popular" middle; decoy dominated or absent.
- All-in total on the card ("99 — total, no fees"); flat over metered; wallet checkout.
- "Why it costs this" block; dated reason on every discount; round=hedonic, precise=utilitarian.

Trust — The 50-Millisecond Verdict

Trust is not evaluated; it is felt in one glance by systems built for faces and threat — then rationalized. Design for the feeling, then survive the fact-check.

5.1 First-impression lock-in and the trust stack

MECHANISM — Visual appeal is judged in ~50 ms and the snap verdict is stable (Lindgaard 2006: 50 ms ratings $\approx$ 500 ms ratings). That affective tag propagates via the **halo effect** into judgments about honesty, competence, and security. Fogg's Prominence-Interpretation theory: whatever has the most visual weight is what gets judged — in his n=2,684 study, **46.1% of all credibility comments were about visual design** (54.6% for finance sites). Prototypicality → fluency → trust: pages that look like a canonical member of their category are believed faster (aesthetic-usability effect, replicated cross-culturally; "ugly converts" is survivorship bias).

DO — First viewport as one composition: one dominant visual, one headline, one CTA, one proof line. Zero layout shift (CLS $\approx$ 0) — reflow reads as "broken" while the verdict forms. ≤ 2 typefaces, one accent, consistent spacing scale. Category-conventional structure (be boring in structure, distinctive in copy). Kill amateurism markers ruthlessly: typos, dead buttons, mixed-quality images, 404s.

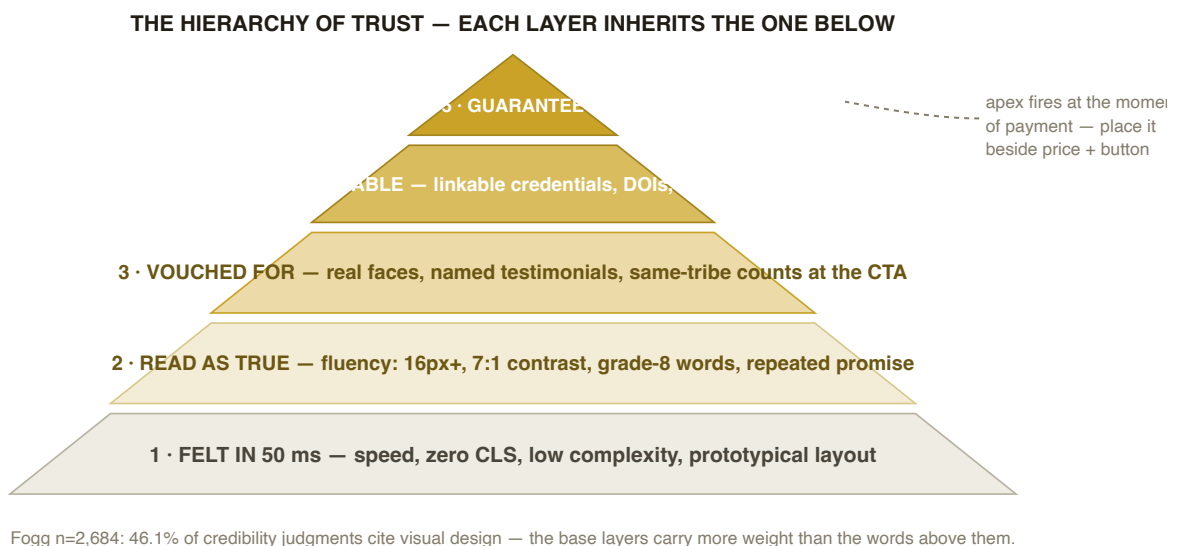


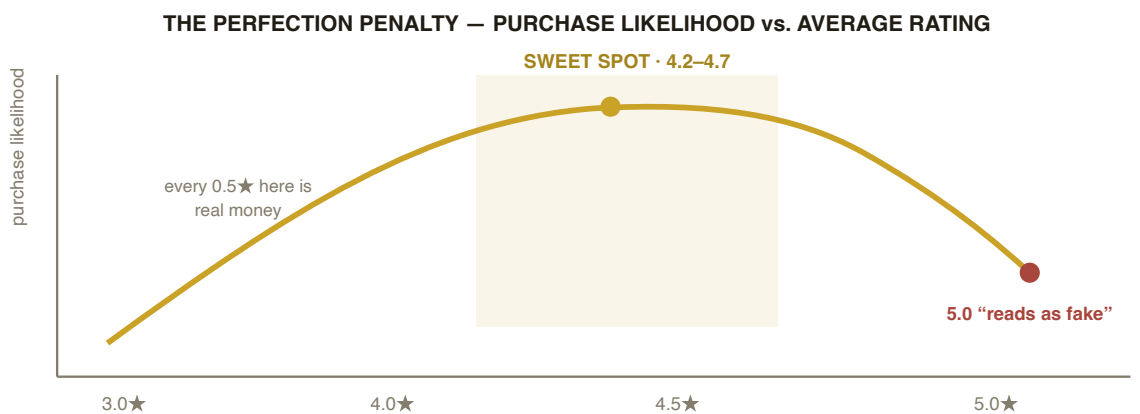
Figure 13 — The hierarchy of trust. A guarantee can't rescue a slow, shifting, ad-cluttered page: each layer only works if the layer beneath it already passed. Audit bottom-up.

5.2 Faces, proof, and optimal imperfection

MECHANISM — Face trustworthiness is computed in **100 ms** (Willis & Todorov), coded implicitly by the amygdala even during unrelated tasks; warmth is judged before competence and weighted more (Fiske). Social proof fires a reinforcement-learning error: disagreeing with the group registers neurally like a lost reward, and the signal's amplitude predicts how far people shift (Klucharev 2009). Perfection backfires: flawless positivity trips **persuasion knowledge** ("I'm being sold to"), while a minor flaw placed *after* positives increases favorability — the blemishing effect (Ein-Gar, Shiv & Tormala 2012).

DO — Real founder/customer photos (a real photo beat the best stock photo by **+35% signups**); full name + role + company under every testimonial; attach the face to the claim it vouches for. Proof at the decision point (rating inside the price card, logos under the CTA), specific and same-tribe ("1,284 freight brokers" > "thousands of users"). Show the full star distribution; add a "who this is *not* for" block after the benefits; state one honest limitation and move on.

EVIDENCE — Spiegel Research Center: reviews lift conversion up to **+270%** (+380% for high-priced items — proof scales with risk); five reviews capture most of the gain; purchase likelihood peaks at **4.2–4.7 stars and falls toward 5.0**; 82% of shoppers actively seek negative reviews; verified-buyer badges $\approx$ +15%.



Spiegel Research Center (Northwestern). Corollary: show the 1–2★ reviews, add a "not for you" block, admit one limitation — after the positives.

Figure 14 — Optimal imperfection. The persuasion-knowledge detector punishes flawlessness. A visible blemish makes the surrounding positives believable — candor is a conversion asset.

5.3 Authority offloads the decision; reassurance can create the fear

MECHANISM — With credible expert input present, value-computation activity in the listener's brain is *attenuated* — the decision is literally offloaded (Engelmann et al. 2009). But specificity is the whole game: unverifiable authority now trips persuasion knowledge. Meanwhile the most-violated law in trust design: risk appraisal is dormant until cued — **your security reassurance is the cue**. "We'll never spam you" on a harmless form *reduces* disclosure (John, Acquisti & Loewenstein 2011).

DO — Make every authority claim checkable: "Co-author, SIGCSE TS 2026 — DOI: 10.1145/3770761.3777339" beats "backed by research." Named humans with photos outperform logos; pair them. Delete "100% safe / no spam" from low-risk forms; security signals **only** where risk is already salient — one recognized mark at the card field. No badge clusters (the NASCAR effect reads as compensating; a McAfee badge has measurably *lowered* conversion).

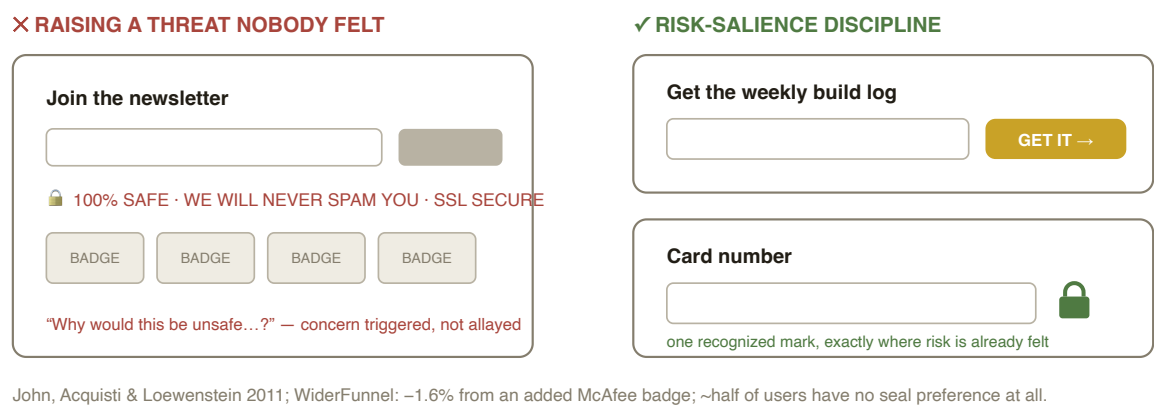


Figure 15 — Silent security. Reassure only where fear already exists. Everywhere else, the reassurance *is* the fear.

TRUST MYTH	REALITY
"Trust badges always lift conversion"	✗ Small effects, sometimes negative; familiarity beats certification strength.
"Ugly sites convert better"	✗ Survivorship bias (Craigslist has lock-in, not a lesson). Aesthetic-usability effect is replicated cross-culturally.
"5.0 stars is the goal"	✗ Peak is 4.2–4.7; perfection reads as fabrication.
"Hard-to-read fonts boost memory"	✗ Failed multi-lab replication (Meyer 2015). Fluency wins; never ship disfluency.
"Reassure users on every form"	✗ Backfires on low-risk forms. Reassure only at the card field.

SHIP LIST — TRUST

- Fold = one composition; CLS $\approx$ 0; prototypical structure; zero amateurism markers.
- Guarantee rewritten concrete ("14 days. One email. Full refund.") and moved beside price + button.
- Real named faces attached to the claims they vouch for; same-tribe counts at the CTA.
- Full review distribution + "not for you" block + one admitted limitation.
- Every authority claim linkable; surplus badges deleted; one lock at the card field.
- Message match: ad headline $\approx$ H1 $\approx$ checkout branding $\approx$ price everywhere.

Memory & Story — Peaks, Endings, and Protagonists

The visitor who leaves doesn't keep your page — they keep a compressed memory of it: the peak, the ending, the first and last things, the concrete things, and the story they were inside.

6.1 Narrative transportation — story switches off counterarguing

MECHANISM — Green & Brock 2000: readers absorbed ("transported") into a story stop generating counterarguments — persuasion runs through an experiential route that bypasses skeptical scrutiny, and the effect persists. The van Laer 2014 meta-analysis (76 papers) confirms it and finds it *stronger* in marketing contexts. Strongest levers: identifiable characters, imaginable plot, verisimilitude — and self-referencing: the reader maps themselves onto the protagonist.

DO — Replace feature dumps with a mini arc: a specific customer who resembles the visitor → concrete struggle → discovery → transformation with real numbers (" +37% in 11 weeks" beats " +40%": rough edges are verisimilitude). Second person: the visitor is the hero, the brand is the guide. Case studies segmented by industry so every visitor meets a protagonist from their world.

EVIDENCE — Significant Objects experiment: ~\$129 of thrift junk resold for **\$3,600+** purely by attaching purpose-written stories — a ~2,700% markup from narrative alone. Beckett Simonon craftsmanship-story panel: +5% sales, 237% annualized ROI.

6.2 Peak-end rule — engineer one peak and a generous ending

MECHANISM — Kahneman's remembering self stores snapshots of the emotional **peak** and the **end**, ignoring duration. Cold-pressor study (Kahneman et al. 1993): 69% — 22 of 32 participants — chose to repeat a *longer* pain because it ended better; the colonoscopy trials replicated it clinically. What your customer remembers — and retells — is the peak moment and the last screen.

DO — Design two moments deliberately: (1) a **peak** — the natural "aha," amplified: a personalized calculator result, a live demo drafting a real reply, a generated report, celebrated visually; (2) the **ending** — a generous, celebratory confirmation page and first email (clear next step, zero upsell-guilt, no exit-intent traps). Treat 404s, errors, and unsubscribes as endings too. The post-purchase memory is what drives repeat purchase and referral.

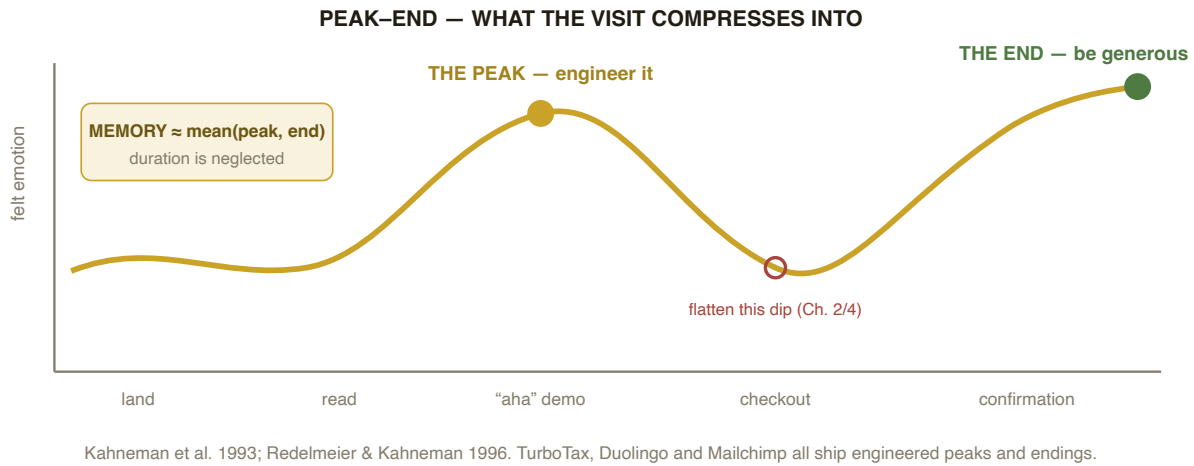


Figure 16 — The remembered visit. Nobody remembers your middle sections. They remember the best moment and the last moment — so build both on purpose, and never end on legalese.

6.3 Serial position — first and last carry the memory

MECHANISM — In any sequence, openings are rehearsed into long-term memory (primacy) and endings sit in short-term memory (recency); the middle gets neither (Ebbinghaus; Murdock 1962; Glanzer & Cunitz 1966 showed the double dissociation). Applies to every list you ship: nav menus, bullet lists, testimonials, tiers, page sections.

DO — Strongest claim opens and closes every list; bury weak points mid-list. Value prop + CTA first on the page (primacy), repeated CTA as the *final* content block (recency). In checkout, end every step on reassurance, not friction.

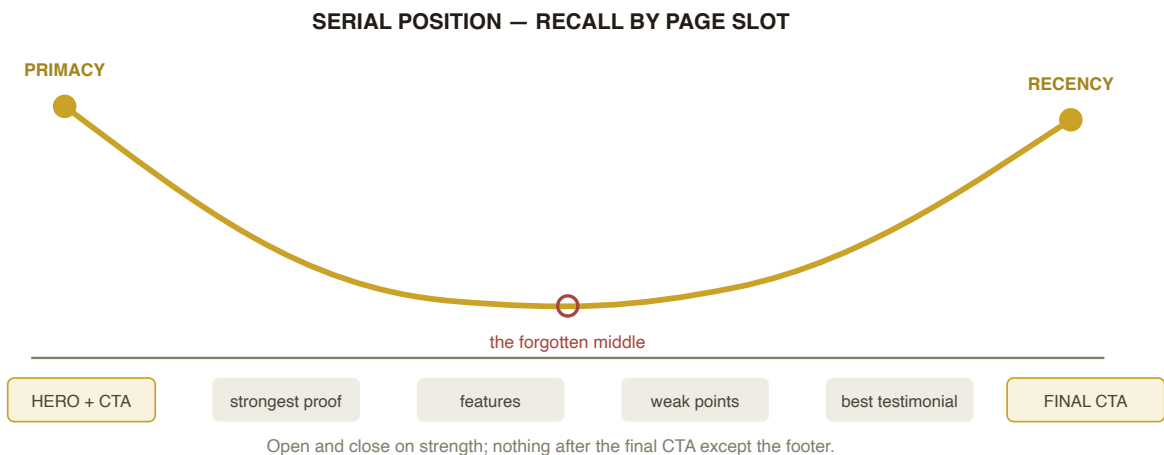


Figure 17 — The U-curve on your page. Position is an argument: the same testimonial is twice as persuasive at the end of the page as in the middle.

6.4 Concrete words, pictured claims

MECHANISM — Paivio's dual coding: concrete words and images get encoded twice (verbal + imagery = two retrieval routes); abstractions get one, and fMRI shows they're literally harder to read. Picture superiority: 24-hour recall was 32% for words, 58% for pictures, **76% for word+picture pairs** — the combo is the winning condition, not the picture alone. NN/g caveat: decorative stock imagery is ignored like a banner ad — only information-carrying images earn fixations.

DO — Rewrite every abstract value prop into a sensory, countable claim: "Streamline your AR workflow" → "Invoices paid in 2 days, not 45." Pair each core claim with a real UI screenshot, workflow diagram, or real-person photo; icon+label pairs in feature grids; testimonials must contain numbers and artifacts. First-person concrete microcopy: "Get my free report," never "Submit."

EVIDENCE — Packard & Berger 2021 (1,000+ real service interactions): concrete language → higher satisfaction, higher purchase, *more actual spend*. Standing, Conezio & Haber 1970: >90% recognition of 2,560 photos days later.

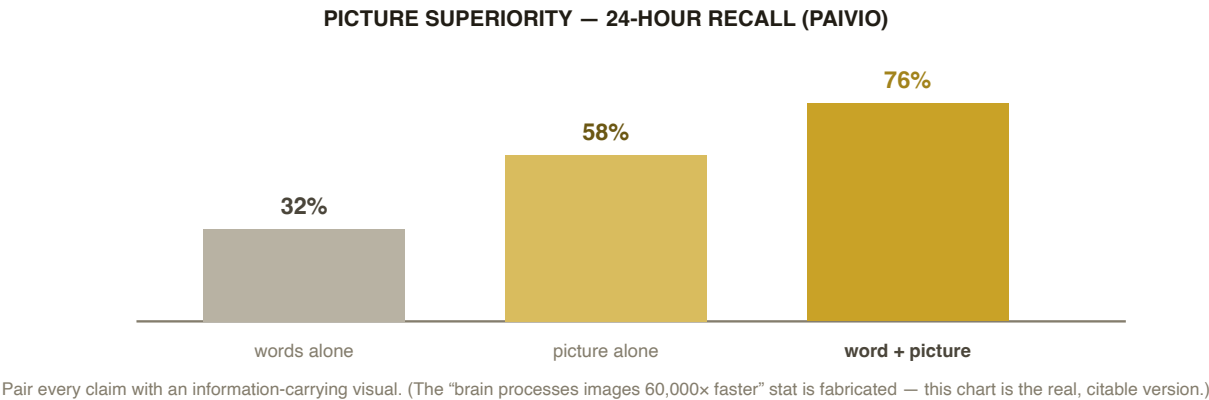


Figure 18 — The 76% condition. Claim + image together more than double the recall of the claim alone. Decorative stock photos don't count — eyes skip them entirely.

MEMORY MYTH	REALITY
"People remember 10% of what they read, 90% of what they do" (learning pyramid)	✗ Fabricated percentages; Dale never wrote numbers on his cone. No primary source exists.
"The brain processes images 60,000x faster than text"	✗ No study exists — traces to a 1982 ad claim. Use picture superiority (76%) instead.
"Stories release oxytocin, the trust molecule"	✗ Zak's findings failed to replicate; measurement methods flawed. The replicated mechanism is transportation / suppressed counterarguing.

SHIP LIST — MEMORY & STORY

- One customer-as-hero arc per selling page, with rough-edged real numbers.
- One engineered peak (calculator, live demo, personalized result) + a celebratory ending.
- Strongest claim first and last in every list; final content block = repeated CTA.
- Every abstract phrase rewritten concrete; every claim paired with a real image.
- Thank-you page and first email treated as the referral engine they are.

Color & Typography — Gold, Dark Luxury, and Fluency

Color meanings are mostly learned and contextual — but contrast, scarcity of accent, and reading geometry are physics. Master the physics; test the hues.

7.1 Color in context: convention first, testing second

MECHANISM — Elliot & Maier's color-in-context theory: the same red means danger on an error, savings on a price tag. Hue→personality associations are learned but real at the brand level: blue → competence/trust, red → excitement/arousal (Labrecque & Milne 2012); saturation and darkness amplify. Shade-level differences convert differently — Google's "41 shades of blue" reportedly moved ~\$200M/yr.

DO — Pick hue by category convention and page job; A/B shades, not meanings; never design from color-meaning infographics. The "85–90% of purchase decisions are color-based" stat is unverifiable — drop it.

7.2 Dark = luxury; gold = scarcity-priced accent

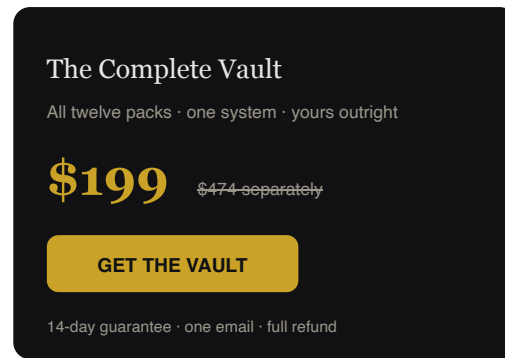
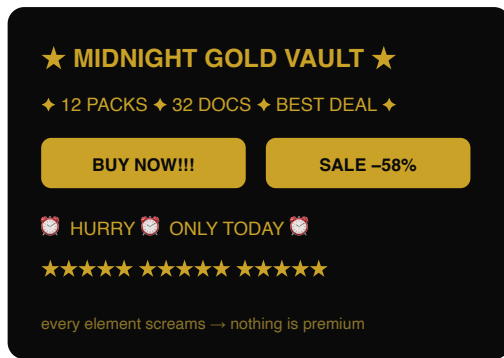
MECHANISM — Black is cross-culturally coded "powerful/expensive" (replicated US/China/Korea/Japan) via luxury-goods association, and dark fields maximize figure-ground pop. Gold inherits the metal's rarity heuristic — but the signal works through **scarcity of use**: gold everywhere reads casino; gold on exactly one element reads premium (Von Restorff again). Dark mode carries a physiological tax: dilated pupils + halation blur light-on-dark text, worst for astigmatic readers at small sizes (Piepenbrock 2013; Dobres 2017).

DO — Near-black #0A0A0A–#141414 (never pure #000), thin weights, desaturated palette, one metallic accent — for brand and landing surfaces. Muted champagne gold (#C9A227 family, never saturated yellow) on *only* the premium tier, badge, or key stat. Pay the dark tax: off-white #E6E6E6 text (not #FFF), body size/weight one step up, no small thin gray-on-black, long-form reading gets a light surface or a toggle.

EVIDENCE — 741-participant packaging study links black elements to perceived premiumness across categories; a pricing-page study found gold tier-coding lifted upgrades ~19%; NN/g's dark-mode research synthesis: light mode wins acuity/proofreading for most sightlines, advantage growing as fonts shrink.

✗ GOLD EVERYWHERE = CASINO

✓ GOLD ONCE = LUXURY



Dark-surface rules: #0A0A0A–#141414 ground (never #000) · #E6E6E6 text (never #FFF) · body one size/weight up · no thin gray-on-black.
The gold budget is the salience budget: spend it on exactly one element per screen.

Figure 19 — The gold budget. Identical palette, opposite signal. Scarcity of accent is what encodes "expensive"; abundance of accent encodes "trying too hard."

7.3 Fluency sells; serif signals; geometry carries

MECHANISM — Processing fluency is misattributed to the *task*: identical exercise instructions in a hard font made people estimate $\sim 2\times$ the effort and drop the intent to do it (Song & Schwarz 2008) — hard-to-read pricing feels hard to use. Typeface associations modulate believability: in Errol Morris's covert NYT experiment ($\sim 45,000$ readers), the same passage was believed most in Baskerville, least in Comic Sans (small but free effect). Reading geometry is saccade mechanics: eyes jump 7–9 letters; long lines cause return-sweep errors, tiny type collapses reading speed below critical print size (Legge).

DO — Serif display for headlines and claims on premium surfaces; humanist sans for UI chrome. Body 16–18 px (18–21 long-form), `max-width:65ch` (45–75 range), `line-height:1.5`, $\geq 4.5:1$ body contrast (7:1 for claims). Nothing condensed, script, or decorative near an action. Never ship deliberate disfluency — the "ugly fonts aid memory" claim (Sans Forgetica et al.) failed replication repeatedly.

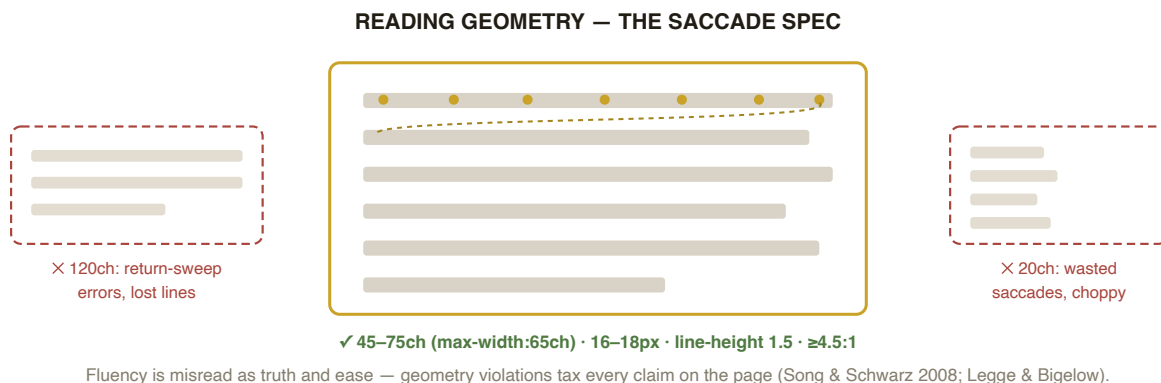


Figure 20 — The saccade spec. 65 characters, 16px+, 1.5 line-height, high contrast. It looks like typography; the brain bills it as credibility.

SHIP LIST — COLOR & TYPE

- One accent hue, reserved for conversion actions, ~6:1+ contrast against its surround.
- Dark luxury surfaces: near-black + off-white + one gold element; long-form gets light or a toggle.
- Serif for claims on premium pages; sans for UI; nothing decorative near an action.
- `max-width: 65ch`, 16–18px body, 1.5 line-height, 4.5:1+ everywhere money moves.

Conversion 2026 — Speed, Thumbs, and the Legal Line

The state of the art, benchmarked: what the biggest datasets (Baymard's 50-study meta, Unbounce's 41k pages, Deloitte/Google's 30M sessions) say actually moves revenue now.

8.1 Speed is a subconscious trust signal — fix it before anything else

MECHANISM — Fluency→trust transfer: delay is processed pre-consciously as friction and taxes every persuasion element downstream. A slow page makes the *offer* feel risky.

DO — Mobile LCP < 2.5 s, interaction < 100 ms; compress/lazy-load images, inline critical CSS, zero blocking third-party JS above the fold. Treat every 100 ms as a revenue line item.

EVIDENCE — Deloitte/Google, 37 brands, 30M+ sessions: **+0.1 s mobile speed → +8.4% retail conversions, +9.2% AOV, +10.1% travel.** Google: bounce probability +32% from 1→3 s. (Retire the 2006 "every second costs 7%" stat — cite this instead.)

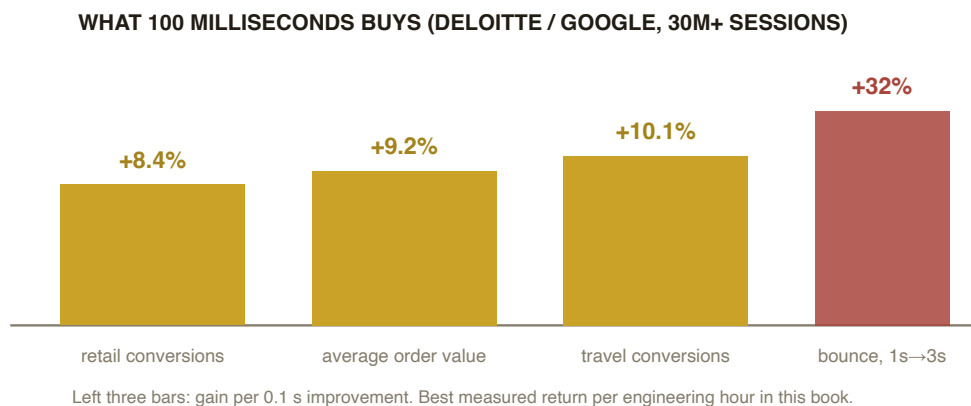


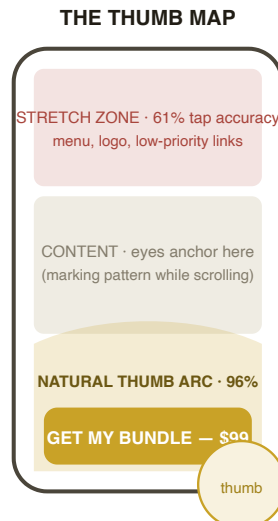
Figure 21 — Speed is persuasion. Milliseconds are a psychological variable: the fastest page feels like the safest vendor.

8.2 Design for the thumb — mobile is where checkout dies

MECHANISM — Fitts on touch: reach cost and error rate explode outside the natural thumb arc. Mobile eyes follow a "marking pattern" — anchored mid-screen while the thumb scrolls — so mid-screen subheads and a fixed bottom action bar match the motor + visual reality.

DO — Primary CTA as a sticky bottom bar in the thumb zone; targets ≥ 44 px (bigger for the buy button); layer-cake subheads for the marking pattern; test flows mobile-first. First-person outcome copy on the button ("Start *my* free trial" beat "your" by +90% clicks — Unbounce/ContentVerve) plus a click-trigger line under it killing the top anxiety ("No card required · 2 min").

EVIDENCE — Thumb-zone accuracy ~96% vs ~61% in the stretch zone; Baymard: mobile cart abandonment **85.65% vs 69.75% desktop** — the gap is the prize. Checkout microcopy "there will be no additional costs" → +11.3%.



CTA ANATOMY — 2026 SPEC

Start my free 30-day trial →

verb + possessive + outcome

No card required · Cancel anytime · Takes 2 min

← click-trigger: kills the top anxiety at the moment of commitment

Why it works:

- "my" > "your": +90% clicks (self-referential encoding)
- outcome-explicit: the click's consequence is predictable
- never "Submit" / "Next" — they name effort, not payoff
- sticky bottom placement: Fitts + marking pattern
- ≥ 44 px tall; the buy button gets more

Mobile abandons 85.7% vs desktop 69.8% (Baymard) — the thumb gap is the

Figure 22 — Thumb-zone conversion. Put the money button where the thumb already rests, written in first-person outcome language, with the anxiety-killer line underneath.

8.3 The composite: before / after

Everything in this book, applied to one landing page. The "before" is not a straw man — it is the median landing page (Unbounce median conversion: 6.6%; most sites sit far below ceiling).

✗ BEFORE — the median page

AUTO-CAROUSEL · 5 messages · slide 2–5 unseen

BUY DEMO TRIAL TOUR CALL

right-rail promos (0.8% of fixations)

wall of abstract text · no subheads · "solutions that streamline synergies"

11 fields incl. "Company (optional)" · account required · price + fees revealed at step 4

SUBMIT

ends on legalese · no guarantee in sight · median fate: ~6.6% conv.

✓ AFTER — the playbook page

GET MY PILOT — \$999

14 days · one email · full refund ← trigger

"1,284 brokers" + 4.6★ (247) — proof at the decision point

subhead every 2–3 ¶ · concrete numbers · word+picture pairs

\$49 ★ MOST POPULAR \$99 \$299

\$285 separately

3 tiers · numeric grid · anchor + compromise · all-in total, no fees

"Not for you if..." — one honest limitation (blemish after positives)

GET MY PILOT →

best testimonial, named + faced

same CTA repeated (serial position) · guest checkout · 7 fields · thumb-zone sticky bar on mobile

Laws applied, left to right: 1–7 (attention), 8–14 (load/fluency), 15–22 (persuasion/pricing), 23–27 (trust), 28–30 (memory/ending).

Figure 23 — Before / after. No single change is dramatic; the compound is. Baymard's checkout meta alone estimates +35% recoverable from design fixes — before touching traffic or product.

PRIORITY ORDER FOR A REAL SITE (HIGHEST LEVERAGE FIRST)

- 1 · Mobile speed (LCP < 2.5s) → 2 · kill surprise costs + guest checkout → 3 · one above-fold CTA with outcome copy + click-trigger → 4 · thumb-zone sticky CTA → 5 · anchored 3-tier pricing → 6 · honest urgency + specific proof → 7 · cosmetic color/contrast tests last.
- Legal line (2025–26): FTC junk-fees rule, CA SB 478, ROSCA enforcement, Amazon's \$2.5B settlement — honest scarcity, all-in pricing, and easy cancellation are now compliance, not just conversion.
- Measurement: most effects land at 2–15%. Detecting +5% at a 3% baseline needs ~50k visitors/variant — below that, ship the strong-prior fixes without testing and save tests for framing questions.

The 30 Laws

The whole book, one number per idea. Cite the source; skip the myth.

01 The 50-Millisecond Verdict

Aesthetic judgment forms in ~50 ms, stays stable, and halos onto trust and usability. Win it with low visual complexity + a prototypical layout.

First impressions · Lindgaard 2006; Tuch/Google 2012 · Ch. 1, 5

02 The Fold Is a Gradient, Not a Wall

57% of gaze above the fold, 74% in two screens. Value prop + CTA in viewport one, no false floors, scent maintained downward.

Information foraging · NN/g 2018; Pirolli & Card · Ch. 1

03 They Scan, They Don't Read

≤28% of words get read. Subhead every 2–3 paragraphs, front-load the first two words, bold key terms — engineer the layer-cake, not the F.

F-pattern/layer-cake · Nielsen 2006/2017; Weinreich 2008 · Ch. 1

04 One Winner Per Screen

Saliency is winner-take-all and zero-sum. One reserved accent hue, one primary button per screenful, whitespace as its surround.

Saliency map / Von Restorff · Itti & Koch 1998 · Ch. 1, 7

05 Don't Look Like an Ad

Ad-shaped, ad-placed, or animated content is suppressed by a learned schema — even when it contains what the user wants.

Banner blindness · Benway 1998; NN/g · Ch. 1

06 Motion Spends Itself Once

Abrupt onsets capture attention only while rare. One entry animation; no loops; no auto-rotating carousels (~1% CTR, 84% slide one).

Onset capture · Yantis & Jonides; Runyon carousel data · Ch. 1

07 **Aim the Eyes You Show**

Faces get privileged processing; averted gaze toward the CTA $\approx 5\times$ product dwell vs. no face. Mutual gaze steals fixations for the face.

Gaze cueing · Sajjacholapunt & Ball 2014 · Ch. 1, 5

08 **Four Chunks of Working Memory**

The real WM limit is $\sim 4 \pm 1$ (not 7). Cut extraneous load; one page, one job; chunk everything under labels; never force carrying info across steps.

Cognitive load · Sweller; Cowan 2001 · Ch. 2

09 **Hick's Law — Fewer, Clearer Choices**

Decision time grows with $\log(N)$ for comparable options. One primary + one ghost CTA; 3 tiers; a preselected default collapses N toward 1.

Choice RT · Hick 1952; Unbounce 41k pages · Ch. 2

10 **Fitts's Law — Big, Close, Thumb-Reachable**

Acquisition time = distance $\div$ size. ≥ 44 px targets, submit adjacent to the last field, sticky bottom bar in the mobile thumb zone.

Motor friction · Fitts 1954; NN/g · Ch. 2, 8

11 **Progressive Disclosure**

Show the primary path now; defer everything secondary behind a click. Guest checkout default; account creation after the receipt.

Disclosure · Nielsen; Baymard · Ch. 2

12 **Every Field Pays Rent**

Each form field costs completion, adds a privacy micro-decision, and invites error. One deleted optional field was worth +\$12M/yr at Expedia.

Form friction · Baymard; Expedia case · Ch. 2

13 **Choice Overload Is Conditional**

The jam study doesn't generalize (meta-mean ≈ 0). Keep the catalog; kill the difficulty: filters, defaults, "Most Popular," comparison grids, low-stakes framing.

Overload moderators · Scheibehenne 2010; Chernev 2015 · Ch. 2

14 Fluency Is Truth

Easy-to-process = judged more true, safer, easier to do. 16px+, high contrast, grade 6–8 words, pronounceable names, pre-computed math, repeated promise verbatim.

Processing fluency · Reber & Schwarz; Song & Schwarz · Ch. 2, 5, 7

15 Speed Is Persuasion

+0.1 s mobile speed → +8.4% retail conversions. LCP < 2.5 s is a trust intervention, not an infrastructure chore.

Fluency→trust · Deloitte/Google 30M sessions · Ch. 8

16 Sell the After-State

Dopamine fires on anticipated reward and migrates to the cue. Show the outcome already achieved; demo > screenshot > bullets; payoff before price.

Wanting vs. liking · Knutson 2001; Berridge · Ch. 3

17 Tame the Insula

Excessive or surprise cost fires a pain-adjacent brake that predicts non-purchase. All-in total early, \$/day reframe, decoupled payment, wallet one-tap.

Pain of paying · Knutson 2007; Prelec & Loewenstein · Ch. 0, 4

18 Anchor Before You Price

The judge computes price – reference; supply the reference first. Real, relevant, precise anchors only — arbitrary anchors failed replication.

Anchoring · Kahneman & Tversky; Janiszewski & Uy · Ch. 4

19 Engineer the Middle

3 tiers + numeric attribute grid + "Most Popular" middle; a dominated decoy and a high anchor push from both sides. Grid required — decoys die without numbers.

Compromise + decoy · Simonson 1989; Huber 1982 · Ch. 4

20 Bundle Losses, Itemize Gains

Integrate every cost into one price; segregate every benefit into its own valued line (\$99 vs. a \$285 itemized stack). Never itemize fees at the decision point.

Hedonic editing · Thaler 1985 · Ch. 4

21 **Justify the Price**

Voluntary cost transparency lifts sales ~21–22%; an unexplained discount breeds quality suspicion. Every discount gets a named, dated reason.

Transparency · Mohan, Buell & John 2020 · Ch. 4

22 **Price Format Follows Buying Mode**

Round prices for hedonic/identity buys; precise for utilitarian/ROI buys; charm endings only when the left digit drops — and never on premium positioning.

Fluency pricing · Wadhwa & Zhang 2015; Thomas & Morwitz · Ch. 4

23 **Proof at the Decision Point**

Deviating from the group registers as a reward error. Specific, same-tribe counts beside the price; ≥5 reviews (+270%, +380% high-ticket); recency beats magnitude.

Social proof · Klucharev 2009; Spiegel · Ch. 3, 5

24 **Optimal Imperfection**

Purchase likelihood peaks at 4.2–4.7 stars and falls toward 5.0. Show the full distribution, add a "not for you" block, admit one flaw — after the positives.

Blemishing effect · Ein-Gar 2012; Spiegel · Ch. 5

25 **Checkable Authority Offloads the Decision**

Credible expert markers literally attenuate the brain's own value computation. Specific + linkable (DOI, named person) or it now reads as a tell.

Authority · Engelmann 2009; Fogg · Ch. 5

26 **Real Faces, Real Names**

Face trust is computed in 100 ms; a real customer photo beat premium stock by +35%. Name, role, company under every testimonial; face beside the claim it vouches for.

Face processing · Willis & Todorov; MarketingExperiments · Ch. 5

27 **Silent Security**

Risk appraisal sleeps until cued — and your reassurance is a cue. No "100% safe" on harmless forms; one recognized mark at the card field only; no badge walls.

Risk salience · John, Acquisti & Loewenstein 2011 · Ch. 5

28 Defaults and Endowed Progress Do the Heavy Lifting

82% vs 42% from a default flip; 34% vs 19% completion from two pre-stamped slots. Preselect the good path; never show an empty progress bar.

Choice architecture · Johnson & Goldstein 2003; Nunes & Drèze 2006 · Ch. 3

29 Concrete Stories Beat Abstract Claims

Transportation switches off counterarguing; concrete, pictured, self-referencing content is remembered at 76% vs 32%. The visitor is the hero; numbers keep rough edges.

Narrative + dual coding · Green & Brock 2000; Paivio; Packard & Berger 2021 · Ch. 6

30 Peak, End, and the Honest Deadline

Memory keeps the peak, the ending, and the first/last items — engineer all three, end every page on the CTA, and use only scarcity you can prove (the FTC now audits the rest).

Peak-end + serial position + reactance · Kahneman; Murdock; FTC 2022–25 · Ch. 3, 6, 8

The Per-Page Checklist

Run every page through the 20 gates, then hit its three highest-leverage laws from the map below. The full page-by-page upgrade spec lives in `neuro-page-spec.json`.

The 20 gates — any page, five minutes

- ☐ **5-second test:** a stranger can state what the page is and what to do next. (Law 1)
- ☐ **Squint test:** blurred, the primary CTA is still the loudest element. (Law 4)
- ☐ Value prop + CTA inside viewport one; next section peeks above the fold. (Law 2)
- ☐ Subheads every 2–3 paragraphs, first two words load-bearing. (Law 3)
- ☐ Nothing ad-shaped, right-railed, or auto-rotating; max one entry animation. (Laws 5–6)
- ☐ Any face on the page looks toward the headline or CTA. (Law 7)
- ☐ Nav $\leq$ 4–5 goal links; exactly one primary action per screen; twin CTAs merged. (Laws 4, 9)
- ☐ Mobile: sticky thumb-zone CTA, $\geq$ 44px targets, first-person outcome copy + click-trigger. (Laws 10, 15)
- ☐ Forms: every field pays rent; guest path default; progress bar starts part-filled. (Laws 11–12, 28)
- ☐ LCP < 2.5s mobile; CLS $\approx$ 0; images compressed and lazy. (Laws 1, 15)
- ☐ Hero shows the after-state, not the mechanism. (Law 16)
- ☐ All-in price visible early; zero surprise line items anywhere. (Law 17)
- ☐ A real anchor sits beside every price, in the same glance. (Law 18)
- ☐ 3 tiers with numeric grid + badged middle wherever plans exist. (Law 19)
- ☐ One price, itemized value stack; costs never partitioned. (Law 20)
- ☐ Every discount carries a named, dated reason; "why it costs this" present. (Law 21)
- ☐ Specific same-tribe proof + guarantee sit beside the price and button. (Laws 23, 17)
- ☐ One honest limitation / "not for you" block after the benefits. (Law 24)
- ☐ Authority claims are linkable; badge clutter deleted; lock at card field only. (Laws 25–26)

- Page ends on the repeated CTA + strongest proof; thank-you page is a celebration. (Laws 29–30)

The nine live pages — top three laws each

PAGE	WHAT IT MUST DO	HIGHEST-LEVERAGE LAWS
store.html	Sell the downloads + bundle	19 Engineer the middle · 20 Itemize the value stack · 23 Proof at the price
freightdesk.html	Book founding pilots	18 Anchor vs. AR clerk · 30 Honest scarcity of slots · 26 Named faces/case proof
templates.html	Sell the \$199 Vault	13 Kill choice difficulty (badge a pack) · 19 Vault as compromise middle · 22 Round premium format
media-kit.html	Win brand deals	18 Rate-card anchor before they name a price · 23 Specific reach proof · 29 Story + before/after pictures
index.html	Route each visitor fast	9 One primary action · 6 Motion discipline · 15 Speed (React shell LCP)
everything.html	One gold map, one next step	9 Hick on five CTAs · 30 End on the strongest door · 10-C Congruence (kill stale "waitlist")
hwyhaul.html	Convert recruiters + feed FreightDesk	29 Case study as story arc · 25 Checkable authority · 30 End on one door, not three
heft.html	Collect founding-100 emails	12 Fix the mailto form · 28 Endowed founding-100 progress · 22 Round luxury pricing
spartaneats.html	Demo the app credibly	1 Frame the demo (prototypicality) · 11 Let visitors browse before signup · 16 After-state first

The compound is the strategy: no law here is worth 50% on its own, but thirty 2–10% edges, stacked on pages that already load fast and tell the truth, is how the median page becomes the outlier.

Sources are cited inline per chapter. Evidence tiers: fMRI/lab mechanism → field experiment → large-scale industry dataset; anything weaker is flagged as vendor-grade. Compiled July 2026 from 8 research syntheses for Yusuf

Gadelrab's Midnight Gold pages.